Toward Calabi–Yau

Extra dimensions in string theory, and what the two-scale-factor notebooks show

Written by Claude (Anthropic) at the request of Bud Simrin · October 2026

Role of AI.

This introduction was written by Claude. The two notebooks it describes, 6D_1 Two Scale Factors and 10D_1, are Bud Simrin's work; the connection between them and Calabi–Yau compactification, and the direction of this section, were suggested by Claude, which also checked the results.

1. Why six curled dimensions?

Superstring theory is consistent only in ten spacetime dimensions. To describe our four-dimensional world, six of them must be curled up into a compact space far too small to see. Which compact space is chosen matters: its shape determines much of the resulting four-dimensional physics. In 1985 Candelas, Horowitz, Strominger and Witten showed that, if some supersymmetry is to survive in four dimensions, the six-dimensional space should be a Calabi–Yau space.

2. What a Calabi–Yau space is

A Calabi–Yau space is a compact space with a complex structure (so that its six real dimensions pair up into three complex ones) and a particular topological property. Calabi conjectured in the 1950s, and Yau proved in 1977, that every such space admits a metric with zero Ricci curvature. Calabi–Yau spaces are therefore not one space but a large class:

- **Complex dimension 1 (2 real):** only the torus T^2, the 'elliptic curve'. The torus of the 6D notebook is the simplest Calabi–Yau space.
- **Complex dimension 2 (4 real):** the torus T^4 and the K3 surface.
- **Complex dimension 3 (6 real):** the 'Calabi–Yau threefolds' used in string theory, of which at least tens of thousands of topologically distinct types are known.

String theorists usually reserve the name for spaces whose curvature is not completely zero (technically, holonomy exactly SU(n)). In that strict sense flat tori are excluded, but for the equations below only Ricci-flatness matters, so a torus and a Calabi–Yau behave the same way.

3. Why no formula for the metric is known

The notebooks find exact solutions because symmetry reduces Einstein's equations to ordinary differential equations in r or t. A compact Calabi–Yau threefold in the strict sense has no continuous symmetries at all, so every component of its metric depends on all six coordinates, and Ricci-flatness becomes a nonlinear partial differential equation (the complex Monge–Ampère equation). Yau's theorem proves that a solution exists but gives no formula. Ricci-flat metrics on Calabi–Yau spaces are instead computed numerically, for example with Donaldson's algorithm (2005), in Headrick and Wiseman's numerical K3 metric (2005), and, since about 2020, with neural networks.

4. Why cosmology does not need the formula

Let ordinary space and the curled space have their own scale factors:

$$ds^2 = c^2\,dt^2 - a(t)^2\,h_3 - b(t)^2\,h_{int},$$

where h_3 is a 3-space of constant curvature k and h_{int} is the metric of a d-dimensional compact 'internal' space. If h_{int} is Ricci-flat, only its Ricci tensor, which is zero, enters the field equations $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. The details of the internal space drop out, and the equations for a(t) and b(t) depend only on d:

$$3\,\ddot{a}/a + d\,\ddot{b}/b = \Lambda c^2$$

$$\ddot{a}/a + 2(\dot{a}/a)^2 + d\,(\dot{a}/a)(\dot{b}/b) + 2kc^2/a^2 = \Lambda c^2$$

$$\ddot{b}/b + (d-1)(\dot{b}/b)^2 + 3\,(\dot{a}/a)(\dot{b}/b) = \Lambda c^2$$

together with the first-order equation (the analog of the Friedmann continuity equation), with D = 4 + d:

$$3(\dot{a}/a)^2 + 3d\,(\dot{a}/a)(\dot{b}/b) + \tfrac{1}{2}d(d-1)(\dot{b}/b)^2 + 3kc^2/a^2 = \tfrac{1}{2}(D-2)\,\Lambda c^2.$$

The notebooks derive these equations from scratch:

- **6D_1 Two Scale Factors (d = 2):** four-dimensional spacetime times a torus T^2, the simplest Calabi–Yau space.
- **10D_1 (d = 6):** four-dimensional spacetime times the six-torus T^6. Because only Ricci-flatness enters, its equations for a(t) and b(t) are exactly those of any Calabi–Yau threefold.

The internal volume is proportional to b^d, so b(t) describes how the size of the curled space changes. In string theory it is called the volume modulus.

5. The stabilization problem

In our universe the curled dimensions, if they exist, must keep a fixed size: their size sets constants of nature such as the strength of gravity in four dimensions, which observations show has not changed measurably. Setting $\dot{b} = \ddot{b} = 0$ in the third equation leaves $0 = \Lambda c^2$. So with only a cosmological constant and a Ricci-flat internal space:

- a frozen internal space is possible only if $\Lambda = 0$; and
- with $\Lambda > 0$, as observed, the internal space must change size.

This is the moduli-stabilization problem, shown here in its simplest form. String theorists address it by adding further ingredients, such as fluxes of higher-dimensional fields, branes, and quantum effects; the best-known proposal is that of Kachru, Kallosh, Linde and Trivedi (2003). Whether such constructions fully succeed is still debated.

6. Explicit Ricci-flat metrics with symmetry

Non-compact Ricci-flat spaces can have symmetry, and then their metrics can be found from ordinary differential equations in one radial variable, the setting of the notebooks.

Eguchi–Hanson (1978).

A four-dimensional Ricci-flat metric, a non-compact Calabi–Yau space of complex dimension 2:

$$ds^2 = dr^2/f + (r^2/4)(\sigma_1{}^2 + \sigma_2{}^2) + (r^2/4)\, f\, \sigma_3{}^2, \quad f = 1 - \rho^4/r^4,$$

where ρ is a constant and $\sigma_1 = \sin\psi\, d\theta - \cos\psi \sin\theta\, d\varphi$, $\sigma_2 = \cos\psi\, d\theta + \sin\psi \sin\theta\, d\varphi$, $\sigma_3 = d\psi + \cos\theta\, d\varphi$ are combinations of the angles of a 3-sphere (Claude checked with SymPy that this metric is Ricci-flat). The $\sigma_3{}^2$ term contains $d\psi\, d\varphi$, so the metric is not diagonal. With ρ = 0 it is flat four-dimensional space. Approximate metrics on the compact K3 surface were built by gluing Eguchi–Hanson patches into a four-torus (Page, 1978).

The conifold (Candelas and de la Ossa, 1990).

A six-dimensional Ricci-flat cone, $ds^2 = dr^2 + r^2\, h_5$, over a particular five-dimensional space h_5, together with smoothed versions whose metrics also follow from radial ordinary differential equations. Such regions appear inside Calabi–Yau threefolds, and they are where many string compactifications concentrate their fluxes.

Either example could be derived with the notebooks' methods, using their support for off-diagonal metrics.

References

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